Pure Mathematics Year 2

Chapter 2

Functions and Graphs

Modulus functions · mappings · domain and range · composite and inverse functions · transformations

Content 2

1. The modulus function

Solve $|3x - 5| = 2 - \frac{1}{2}x$.

2. Mappings, domain and range

If $f(x) = x^2 - 4x + 3$, find the range of f.

3. Composite functions

For $f(x) = 2x + 1$, $g(x) = x^2$, determine $fg(x)$ and $gf(x)$.

4. Inverse functions

If $f(x) = \dfrac{x + 1}{2x - 1}$, determine $f^{-1}(x)$.

5. Transforming graphs

Sketch $y = |f(x)|$ and $y = f(|x|)$.

Combine transformations and transform the modulus function.

The modulus of a number

The modulus of a, written $|a|$, is its non-negative numerical value.

$|6| = 6, \qquad |-7.1| = 7.1.$

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Distance and magnitude

The distance between two numbers a and b is $|b - a|$, regardless of their order.

For example, the distance between 2 and 7 is

$|7 - 2| = |2 - 7| = 5.$

Modulus measures size: distances and speeds are non-negative. The same notation $|\mathbf{a}|$ gives the magnitude (length) of a vector.

Examples 4

If $f(x) = |2x - 3| + 1$, find:

(a) $f(5)$

a

(b) $f(-2)$

b

(c) $f(1)$

c

Sketch $y = |x|$

Complete the values, then sketch the graph.

$x = -2$	$x = -1$	$x = 0$	$x = 1$	$x = 2$
1	2	3	4	5

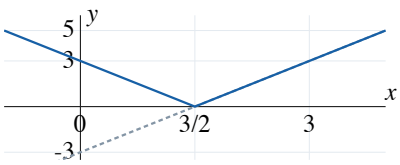
Graph

How to sketch $y = |ax + b|$

Sketch $y = ax + b$. Keep the part on or above the x-axis and reflect the part below the axis upwards.

Every output of a modulus function is non-negative.

1. Sketch $y = |2x - 3|$



The dotted line is $y = 2x - 3$.
Reflect its negative part upwards.
The vertex is $(\frac{3}{2}, 0)$.

2. Solve $|2x - 3| = 5$

2: Graph

2: Solutions

3. Solve $|3x - 5| = 2 - \frac{1}{2}x$

3: Graph

3: Solutions

4. Solve $|3x - 5| > 2 - \frac{1}{2}x$

4

Test Your Understanding 7

1. Solve $|x + 1| = 2x + 5$. There is only one solution.

1: Graph

1: Working

2. Solve $|4x - 1| < 2x$.

2: Graph

2: Working

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5. $g(x) = |4 - \frac{3}{2}x|$, $h(x) = 5$.

(a) Sketch both graphs on the same axes.

(b) Hence solve $|4 - \frac{3}{2}x| = 5$.

5a

5b

8. Solve $|3x - 5| = 11 - x$.

8: Graph

8

9. (a) On the same axes, sketch $y = |6 - x|$ and $y = \frac{1}{2}x - 5$.

(b) State, with a reason, whether $|6 - x| = \frac{1}{2}x - 5$ has any solutions.

9a

9b

12. Solve $|2x + 9| < 14 - x$.

12: Graph

12

13. The equation $|6 - x| = \frac{1}{2}x + k$ has exactly one solution.

(a) Find k. (2 marks)

(b) State the solution. (2 marks)

13: Graph

13a

13b

What is a mapping? 10

What is a mapping?

A mapping relates a set of inputs to a set of outputs. It may follow a rule or be described by individual arrows.

The **domain** is the set of possible inputs. The **range** is the set of outputs that are actually reached.

An arbitrary mapping

$0 \mapsto 4.4$ and 7.2, $2 \mapsto 1$, $3.1 \mapsto 7.2$.

One input can relate to several outputs; different inputs can share an output. Some listed inputs may have no arrow.

Example: $x \mapsto 2x$

x	-1	0	1.7	2
$2x$	-2	0	3.4	4

A general relation may have more than one output for an input, or no output for some listed inputs. A function has a stricter rule, explored next.

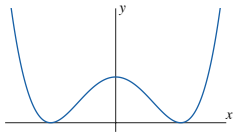
What is a function? 11

A function maps every element of its domain to exactly one output.

Notation: $f(x) = 2x + 1$ or $f : x \mapsto 2x + 1$. Here $f(x)$ denotes the output for input x.

A graph represents the points $(x, f(x))$. The vertical line test checks whether an input has more than one output.

A. This graph:

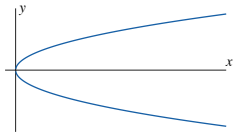


Yes

No

A: Reason

B. This graph:



Yes

No

B: Reason

C. $f(x) = 2^x$

Domain: $\mathbb{R}$.

Yes

No

C: Reason

D. $f(x) = \sqrt{x}$

Domain: $\mathbb{R}$.

Yes

No

D: Reason

E. $f(x) = \pm\sqrt{x}$

Domain: $x \geq 0$.

Yes

No

E: Reason

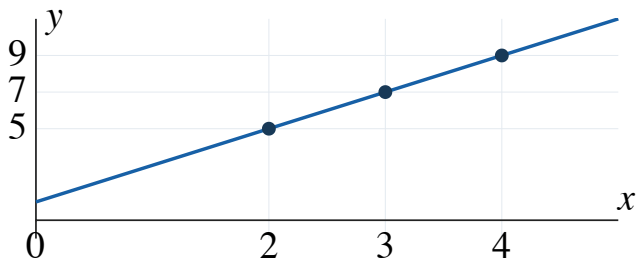
One-to-one vs Many-to-one 12

Each input of a function has one output. Different inputs may still share the same output. The horizontal line test distinguishes these two types.

One-to-one

Every output in the range comes from exactly one input.

Example: $f(x) = 2x + 1$.

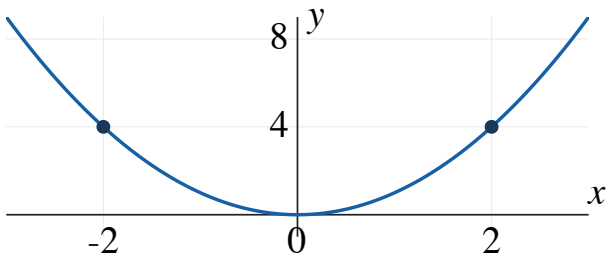


One-to-one

Many-to-one

Two or more inputs can have the same output.

Example: $f(x) = x^2$, $x \in \mathbb{R}$.



Many-to-one

Further Examples 13

Find each range and state whether the function is one-to-one or many-to-one. Use a sketch where helpful. Write ranges in terms of the output, not the input.

(a) $f(x) = 3x - 2$
Domain: $\{1, 2, 3, 4\}$.

a

(b) $g(x) = x^2$
Domain: $-5 \le x \le 5$.

b: Sketch

b: Range

(c) $h(x) = 1/x$
Domain: $0 < x \le 3$.

c: Sketch

c: Range

Piecewise Functions 14

A **piecewise function** uses different rules on different parts of its domain.

$$f(x) = \begin{cases} 5 - 2x, & x < 1, \\ x^2 + 3, & x \ge 1. \end{cases}$$

(a) Sketch $y = f(x)$ and state the range.

(b) Solve $f(x) = 19$.

a: Sketch

a: Range

b

Edexcel C4 June 2012, Q6(a)

$f(x) = e^x + 2, x \in \mathbb{R}$. State the range of f.

1: Sketch

1: Range

Edexcel C4 June 2010, Q4(d)

$g(x) = x^2 - 4x + 1, 0 \leq x \leq 5$. Find the range.

2: Sketch

2: Range

7. The function s is defined by

$$s(x) = \begin{cases} x^2 - 6, & x < 0, \\ 10 - x, & x \geq 0. \end{cases}$$

(a) Sketch $y = s(x)$.

(b) Find all a such that $s(a) = 43$.

(c) Solve $s(x) = x$.

7a

7b

7c

8. The function p is defined by

$$p(x) = \begin{cases} e^{-x}, & -5 \le x < 0, \\ x^3 + 4, & 0 \le x \le 4. \end{cases}$$

(a) Sketch $y = p(x)$.

(b) Find all a, to 2 decimal places, such that $p(a) = 50$.

8a

8b

The left branch ends at the open point $(0, 1)$; the right branch starts at the filled point $(0, 4)$.

9. The function h has domain $-10 \le x \le 6$. It is linear from $(-10, 14)$ to $(-4, 2)$, and from $(-4, 2)$ to $(6, 27)$.

(a) Sketch $y = h(x)$.

(b) State the range.

(c) Find all a such that $h(a) = 12$.

9a

9b

9c

11. $f(x) = ax^3 + bx - 5$, where a, b are constants. Given $f(1) = -4$ and $f(2) = 9$, find a, b.

11

12. $h(x) = x^2 - 6x + 20$, with domain $x \geq a$. Given that h is one-to-one, find the smallest possible a. (6 marks)

12: Sketch

12

Summary of Domain/Range 20

Identify the range of each function. Sketch the graph and check whether the domain is restricted.

1. $f(x) = x^2$, $x \in \mathbb{R}$.

1

2. $f(x) = 1/x$, $x \neq 0$.

2

3. $f(x) = \ln x$, $x > 0$.

3

4. $f(x) = e^x$, $x \in \mathbb{R}$.

4

5. $f(x) = x^2 + 2x + 9$, $x \in \mathbb{R}$.

5

6. $f(x) = x^2$, $-1 \leq x \leq 4$.

6